Source:

xWalker - Enhanced Inspection and Manipulation for Lunar Assembly, NASA:

https://techport.nasa.gov/projects/158648

Product Summary:

It is an advanced autonomous robotic framework that is designed to support the large-scale infrastructure on the lunar surface. This robot uses specialized locomotion capabilities to step across, climb or walk along structural lattices, allowing it to inspect or add building frames. It also has high dexterity manipulation allowing it to grasp complex objects and handle sensitive payloads with extreme precision. Overall, the xWalker is designed to perform autonomous assembly, outfitting and inspection behaviors in extreme space environments like the Lunar Surface.

Product Critique:

The xWalker can only lift up to 10 kg of payload, so it can only carry a small amount of load. While the 7 DOF systems provide excellent dexterity, its small structural footprint restrains it from being able to reach deployable bones or crane based manipulation systems. Our robot will have to be able to carry much heavier and larger loads while implementing a crane like robotic arm to carry this load.

Images/Sketches:



